LOGISTICS OPERATIONS & FLEET PERFORMANCE ANALYTICS

Project Report

Project Type: Logistics Analytics Portfolio Project
Analysis Period: January 2022 – December 2024
Tools: MySQL | SQL | Power BI | DAX | Power Query | Microsoft Word

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PROJECT OVERVIEW

This project analyses logistics operations and fleet performance using multiple related datasets covering transportation loads, trips, customers, routes, drivers, trucks, trailers, fuel purchases, delivery events, facilities, maintenance and safety incidents.

The objective is to demonstrate how SQL-based analysis, data modelling and business intelligence visualisation can be used to evaluate operational performance, identify service-level issues, assess fleet-related expenditure and highlight opportunities for further investigation. The project combines relational data validation and SQL analysis with an interactive Power BI dashboard. The findings are presented through an analytical report that translates operational data into business insights and practical recommendations.

The analysis covers three principal areas:

- **Service and commercial performance:** Delivery reliability, detention, route performance, revenue and revenue per mile.
- **Fleet performance:** Fuel expenditure, fuel efficiency, maintenance costs and vehicle utilisation.
- **Data quality and decision support:** Record completeness, relationship validation, metric limitations and opportunities for more reliable reporting.

The project is designed to demonstrate practical analytical skills relevant to junior logistics analyst, supply chain analyst, transport analyst and fleet performance analyst roles.

BUSINESS PROBLEM

Logistics organisations generate data across multiple business processes, including transportation planning, delivery execution, fleet maintenance, fuel purchasing and driver performance.

However, individual datasets do not always provide management with an integrated view of operational performance. Without appropriate data validation and analysis, it can be difficult to identify service-level weaknesses, understand changes in expenditure or determine which operational areas require further investigation.

This project addresses the following business question:

How can a logistics operation improve service reliability and asset efficiency while maintaining revenue performance?

The investigation focuses on the following questions:

1. How effectively is the operation generating revenue relative to transportation activity?
2. What proportion of deliveries are completed on time?
3. Do delivery reliability and detention vary meaningfully between routes and facilities?
4. What explains changes in recorded fuel expenditure between 2022 and 2024?
5. Which trucks account for the highest maintenance expenditure?
6. How does recorded truck utilisation vary across the fleet?
7. Can comparing maintenance expenditure and utilisation help identify vehicles requiring further investigation?
8. What data quality issues should management consider before making operational decisions?

The analysis is descriptive and diagnostic. It identifies patterns and areas for investigation rather than assuming that observed relationships establish causation.

PROJECT OBJECTIVES

The objectives of this project are to:

- Understand the structure and relationships of the available logistics datasets.
- Validate record counts, primary keys, dates and important relationships.
- Identify missing identifiers, duplicate records and potential data quality limitations.
- Use SQL to investigate operational performance and calculate relevant metrics.
- Develop an appropriate relational data model for Power BI reporting.
- Create business performance measures using DAX.
- Develop an interactive dashboard covering commercial, service and fleet performance.
- Identify meaningful operational patterns and potential improvement opportunities.
- Communicate the findings through a professional analytical report.
- Demonstrate the application of data analysis to practical logistics and supply chain questions.

DATA AND METHODOLOGY

Dataset overview:

The project uses 14 related datasets representing different aspects of logistics operations.

Dataset	Grain	Primary Key	Important Relationships
Loads	One transportation load per row	load_id	customer_id, route_id
Trips	One trip record per row	trip_id	load_id, driver_id, truck_id, trailer_id
Customers	One customer per row	customer_id	Loads
Drivers	One driver per row	driver_id	Trips, driver monthly metrics, safety incidents
Trucks	One truck per row	truck_id	Trips, fuel purchases, maintenance records, utilisation metrics
Trailers	One trailer per row	trailer_id	Trips
Routes	One route per row	route_id	Loads

Fuel Purchases	One fuel transaction per row	fuel_purchase_id	Truck, driver and trip identifiers where available
Delivery Events	One pickup or delivery event per row	event_id	load_id, trip_id and facility_id where available
Facilities	One facility record per row	facility_id	Delivery events
Maintenance Records	One maintenance event per row	maintenance_id	truck_id
Safety Incidents	One incident per row	incident_id	trip_id, truck_id and driver_id where available
Driver Monthly Metrics	One driver-month record per row	Composite driver-month key	driver_id
Truck Utilisation Metrics	One truck-month record per row	Composite truck-month key	truck_id

Note: The actual dataset field names and available relationships should be confirmed against the database schema. Monthly metric tables require a composite identifier, such as driver and month or truck and month, rather than the individual driver or truck identifier alone.

The dataset includes operational records spanning 2022 - 2024. Some fuel purchase and delivery event timestamps extend into early January 2025. The principal reporting period is therefore restricted to 1 January 2022 through 31 December 2024 where the relevant visuals and measures support this filter.

Tools and workflow:

The project followed a structured analytical workflow.

Stage 1: Data inspection and validation

The CSV datasets were imported into MySQL. Record counts, key uniqueness, date ranges, missing identifiers and selected business rules were checked to assess whether the data was suitable for analysis.

Stage 2: SQL analysis

SQL queries were used to investigate delivery reliability, route performance, detention, revenue, fuel expenditure, fuel consumption, maintenance costs and fleet utilisation.

Stage 3: Data modelling

The datasets were connected in Power BI using their relevant identifiers. Relationships were reviewed to support reporting across loads, trips, routes, facilities and fleet-related tables.

Stage 4: KPI development

DAX measures were created for metrics including total revenue, total loads, total miles, revenue per mile, on-time delivery percentage, average delivery detention and total fuel expenditure.

Stage 5: Dashboard development

Power BI was used to present the results through three pages: Executive Overview, Service & Route Performance, and Fleet & Driver Performance.

Stage 6: Interpretation and recommendations

The results were assessed against the business problem. Recommendations were developed while distinguishing confirmed findings from hypotheses that require additional investigation.

Key analytical definitions:

Total revenue

Calculated as the sum of load revenue, fuel surcharge and accessorial charges.

Revenue per mile

Calculated by dividing total revenue by total actual trip miles. This indicates revenue generated per recorded mile but does not measure profitability.

On-time delivery percentage

Calculated as the number of delivery events recorded as on time divided by all delivery events classified as deliveries.

Average delivery detention

Calculated using detention records for delivery events only, rather than combining pickup and delivery events.

Fuel expenditure

Calculated as the sum of recorded fuel purchase costs. This measures recorded expenditure and is not, by itself, a measure of fuel efficiency.

Fleet fuel efficiency

Assessed using the available annual miles-per-gallon results. This metric is distinct from fuel expenditure because expenditure can change when fuel prices change.

Average truck utilisation rate

Calculated as the average of recorded utilisation rates for each truck. These rates require careful interpretation because some source records exceed 100%.

DATA QUALITY ASSESSMENT

Dataset validation:

The imported data contains the following principal record counts:

Dataset	Records
Loads	85,410
Trips	85,410
Delivery Events	170,820
Fuel Purchases	196,442
Maintenance Records	2,920
Driver Monthly Metrics	4,464
Truck Utilisation Metrics	3,312
Drivers	150
Trucks	120
Trailers	180
Customers	200
Routes	58
Facilities	50
Safety Incidents	170

The loads table contains 85,410 unique load identifiers, with no duplicate load identifiers detected. All load records were marked as completed, and the load dates covered 1 January 2022 to 31 December 2024.

The trips table also contains 85,410 unique trip records. The analysis confirmed one trip per load in the supplied data.

The delivery events table contains two event records per load overall: 85,410 pickup events and 85,410 delivery events. Delivery performance calculations therefore need to filter for delivery events to avoid including pickup events in the denominator.

Missing identifiers:

The trips dataset contains missing operational identifiers:

Field	Missing Records	Approximate Share
truck_id	1,672	1.96%
driver_id	1,714	2.01%
trailer_id	1,680	1.97%

These records limit the ability to attribute every trip to a specific truck, driver or trailer.

The analysis did not identify unmatched nonblank identifiers in the relevant checks; the principal issue was missing identifiers rather than references to nonexistent entities.

The safety incidents dataset also contains one missing truck identifier and one missing driver identifier. These omissions may limit the ability to link individual incidents to fleet or driver performance.

Utilisation data quality:

The truck utilisation metrics table contains 3,312 truck-month records, with no duplicate truck-month combinations identified. However, 436 records, approximately 13.16% of the table, have utilisation rates above 100%. The observed rates range from approximately 0.387 to 1.484. Values above 100% may reflect the dataset's calculation method, operational assumptions or data quality issues. The supplied data does not establish which explanation is correct.

Consequently, utilisation rates are retained for descriptive comparison but should not be treated as independently validated measures of physical fleet capacity. The report recommends confirming the calculation and expected range before using the metric for operational targets.

Fuel and delivery date boundaries:

Some fuel purchase records extend into early January 2025, beyond the core reporting period. The principal reporting period is January 2022 to December 2024. Records outside this period should be excluded from comparisons intended to cover the three complete calendar years. Delivery event records also extend into early January 2025. Date filters must therefore be applied consistently when comparing delivery and fleet performance over the defined period.

Detention data limitations:

The maximum recorded detention is 239 minutes, and the same maximum appears across all facilities. This uniformity may indicate a recording cap or a characteristic of the dataset. The available information does not confirm the reason. The maximum should therefore be investigated before it is interpreted as a comparable real-world extreme across facilities. Average detention remains useful for descriptive comparison, subject to the limitations of the underlying data.

Data quality conclusion:

The data was sufficiently structured to support the selected descriptive analyses and dashboard KPIs. However, missing trip identifiers, utilisation rates above 100%, the repeated maximum detention value and records outside the reporting period require careful handling. These limitations do not invalidate every finding, but they affect the level of confidence that can be placed in certain comparisons. The dashboard should be interpreted as a performance analysis of the supplied dataset, rather than a fully reconciled operational reporting system.

ANALYSIS AND FINDINGS

Delivery reliability:

The analysis identified 85,410 delivery events within the core reporting period.

Metric	Result
On-time deliveries	38,102
Late deliveries	47,308
On-time delivery percentage	44.61%
Late delivery percentage	55.39%
Average delivery detention	106.60 minutes
Maximum recorded detention	239 minutes

Only 44.61% of deliveries were recorded as on time, while 55.39% were recorded as late.

This indicates a significant service reliability concern within the supplied dataset. The result warrants further investigation into scheduling, transit times, dispatch practices, facility processing, customer receiving windows and the accuracy of recorded delivery timestamps.

The average delivery detention of 106.60 minutes also indicates that recorded delivery events involve substantial time spent waiting or delayed at delivery facilities, depending on the dataset's definition of detention. However, the available data does not establish that detention alone causes late deliveries. Further analysis should examine the relationship between detention, planned arrival times, actual arrival times, route characteristics and delivery outcomes.

Recommendation: Establish a service improvement baseline and investigate the principal contributors to late deliveries. Prioritise the operational causes supported by additional evidence rather than assuming a single cause.

Route performance:

Route-level analysis revealed differences in on-time delivery performance and revenue generated per mile.

Measure	Route	Result
Lowest on-time delivery percentage	RTE00029	42.40%
Highest on-time delivery percentage	RTE00021	48.00%
Highest revenue per mile	RTE00015	3.62
Lowest revenue per mile	RTE00043	1.70

The difference between the highest and lowest route-level on-time percentages is relatively modest compared with the overall network's late delivery rate.

This suggests that the service reliability problem is not confined to one isolated route. The variation between routes still warrants investigation, but the broader network performance should remain a priority. The revenue-per-mile results also demonstrate that routes differ in the revenue generated relative to recorded distance travelled.

However, higher revenue per mile does not necessarily indicate higher profitability. The analysis does not include all costs required to determine route-level profit, such as driver compensation, insurance, tolls, depreciation and other operating overheads.

Recommendation: Investigate the scheduling, distance, service requirements and operating characteristics of lower-performing routes. Review revenue per mile alongside available cost and service data before making route allocation or pricing decisions.

Facility performance and detention:

Facility analysis identified the following results:

Facility	Facility Name	Average Detention
FAC00004	Chicago Distribution Center	109.62 minutes
FAC00024	Denver Distribution Center	109.22 minutes
FAC00015	Phoenix Distribution Center	109.12 minutes
FAC00048	Indianapolis Warehouse	108.88 minutes
FAC00012	Atlanta Distribution Center	108.85 minutes

The facilities with the lowest recorded average detention included:

Facility	Facility Name	Average Detention
FAC00036	Omaha Distribution Center	101.57 minutes
FAC00034	Charlotte Terminal	102.57 minutes
FAC00033	Miami Distribution Center	103.25 minutes
FAC00047	Milwaukee Hub	104.50 minutes
FAC00005	Indianapolis Hub	104.81 minutes

The results show differences in average detention between facilities. However, lower detention does not consistently correspond to a higher on-time delivery percentage.

This means facility detention should be assessed alongside other factors, including scheduled arrival times, traffic, loading and unloading processes, delivery appointment adherence and route characteristics. Facility identifiers were used to distinguish records because facility names are not unique in the supplied dataset.

Recommendation: Investigate the facilities with the highest average detention and compare their operating processes with lower-detention facilities. Use the results to identify potentially transferable practices, while accounting for differences in facility function and delivery volume.

Revenue and transportation activity:

The analysis produced the following overall measures:

Metric	Result
Total revenue	Approximately 298.62 million recorded currency units
Total loads	85,410
Total actual miles	Approximately 122.2 million
Revenue per mile	Approximately 2.44 recorded currency units

The results provide a high-level view of the relationship between recorded revenue and transportation activity.

Revenue per mile provides a consistent way to compare revenue generation relative to distance travelled. However, it does not account for the complete cost of providing the transportation service. The source data does not establish a confirmed currency denomination. Accordingly, values are presented as recorded currency units rather than being labelled as a specific currency.

Recommendation: Use revenue per mile as an initial commercial performance indicator, then incorporate fuel, maintenance, driver, insurance, toll and other relevant costs before evaluating route profitability.

Fuel expenditure analysis:

Recorded fuel expenditure declined between 2022 and 2024.

Year	Fuel Expenditure	Fuel Consumption	Recorded Cost per Gallon
2022	34,345,178.89	8,181,816.3 gallons	4.198
2023	31,073,119.57	8,070,497.5 gallons	3.850
2024	30,081,424.68	8,241,247.0 gallons	3.650

Between 2022 and 2024:

- Recorded fuel expenditure declined by approximately 12.4%.
- Recorded cost per gallon declined by approximately 13.1%.
- Fuel consumption increased by approximately 0.7%.

The reduction in expenditure occurred alongside a decline in recorded fuel cost per gallon, while fuel consumption remained broadly stable.

The results therefore suggest that lower recorded fuel prices, rather than a substantial reduction in fuel consumption, were the principal factor associated with the decline in fuel expenditure. This distinction is important because a reduction in fuel spending does not automatically indicate an improvement in fleet efficiency.

Recommendation: Monitor fuel expenditure and fuel consumption separately. Combine fuel purchase records with validated mileage data to assess vehicle-level fuel efficiency and identify potential opportunities for improvement.

Fleet fuel efficiency:

The annual fleet MPG results were:

Year	Fleet MPG
2022	6.444
2023	6.445
2024	6.453

The fleet's recorded MPG improved by approximately 0.14% between 2022 and 2024.

This represents a marginal improvement in fuel efficiency over the reporting period. When considered alongside the reduction in fuel expenditure and recorded cost per gallon, the findings indicate that fuel-price changes were a more significant factor in reduced fuel spending than the measured improvement in MPG. The fleet MPG results should be treated as an annual summary of the available data. Further validation is required before attributing the observed change to specific drivers, trucks, maintenance practices or operational decisions.

Recommendation: Investigate truck-level MPG, excessive idling, driving behaviour, vehicle condition and maintenance history. Use consistent mileage and fuel-consumption measures to assess whether targeted interventions produce measurable improvements.

Maintenance cost analysis:

Maintenance records were analysed to identify trucks with high recorded maintenance expenditure and to distinguish maintenance frequency from average cost per event.

The ten highest-total-cost trucks included the following:

Truck ID	Maintenance Events	Total Maintenance Cost	Average Cost per Event
TRK00003	41	90,161.42	2,199.06
TRK00073	39	76,573.99	1,963.44
TRK00081	32	73,892.72	2,309.15
TRK00099	34	73,231.77	2,153.88
TRK00014	34	71,801.82	2,111.82
TRK00075	30	71,073.64	2,369.12
TRK00118	35	71,005.77	2,028.74
TRK00026	32	70,708.48	2,209.64
TRK00044	38	69,825.23	1,837.51
TRK00038	31	68,989.14	2,225.46

TRK00003 recorded the highest total maintenance expenditure among these trucks, at 90,161.42 across 41 maintenance events. TRK00075 had the highest average cost per maintenance event among the ten listed trucks, at 2,369.12. TRK00044 recorded 38 events but had a lower average cost per event of 1,837.51.

These results demonstrate why total maintenance expenditure, event frequency and average repair cost should be examined separately. A truck may have high total costs because it experiences frequent maintenance, expensive repairs or a combination of both. The analysis does not establish whether these vehicles are unreliable or uneconomical to operate. Their mileage, age, utilisation, downtime and maintenance types must also be considered.

Recommendation: Investigate high-cost trucks by comparing maintenance expenditure with mileage, vehicle age, utilisation and downtime. Prioritise vehicles where several independent indicators suggest a persistent maintenance or reliability concern.

Fleet utilisation analysis:

The fleet utilisation visual compares the average recorded utilisation rate of trucks, with the chart restricted to the ten highest-average trucks. TRK00055 recorded the highest average utilisation rate among the trucks displayed, at approximately 0.89, equivalent to 89% when the source rate is interpreted as a decimal.

This comparison provides a view of the trucks with the highest recorded average utilisation. However, the source dataset includes individual monthly rates above 100%, which means the metric's calculation and interpretation should be confirmed. The top-ten visual is useful for highlighting differences between trucks, but a high recorded utilisation rate does not automatically establish that a truck is optimally utilised. The operational meaning of the measure, the vehicle's available capacity and the demand placed on it must also be considered.

Recommendation: Validate the utilisation calculation and investigate trucks with unusually high or low rates. Use confirmed utilisation data alongside mileage, maintenance expenditure, downtime and service performance to inform fleet planning.

Comparing maintenance cost and utilisation:

The Power BI dashboard presents two complementary fleet visuals:

- Top 10 trucks by total maintenance cost.
- Top 10 trucks by average utilisation rate.

Displaying these charts side by side makes it easier to compare the trucks requiring attention for maintenance expenditure with those recording the highest average utilisation. This comparison is intended to support further investigation into whether trucks with high utilisation also incur substantial maintenance costs.

However, the two charts rank trucks using different measures. A truck appearing in both top-ten lists does not, by itself, establish that utilisation caused higher maintenance costs. Similarly, a truck appearing in only one list is not necessarily performing poorly. A stronger assessment would compare the same trucks using maintenance cost per mile, maintenance events per mile, downtime, vehicle age and utilisation over a consistent period.

Recommendation: Identify trucks appearing in both top-ten groups, then investigate their mileage, age, maintenance history and downtime. Where reliable data is available, calculate maintenance cost per mile and compare similar vehicles before recommending changes to fleet allocation or replacement decisions.

POWER BI DASHBOARD OVERVIEW

The Power BI dashboard translates the analysis into three pages designed to support different management questions.

Page 1 - Executive Overview:

Purpose: Provide a high-level summary of commercial performance, transportation activity and delivery reliability.

The page includes KPI cards for total revenue, total loads, total miles, revenue per mile, on-time delivery percentage and average detention. A monthly revenue trend provides visibility into changes in recorded revenue over time. A booking-type visual compares load volumes across booking categories, while the delivery timeliness breakdown shows the proportion of on-time and late deliveries.

The principal insight is that the operation generated approximately 298.62 million recorded currency units in revenue across 85,410 loads, while only 44.61% of deliveries were recorded as on time. This page gives users a starting point for understanding the scale of the operation and identifying the service reliability issue requiring further investigation.

Page 2 - Service & Route Performance:

Purpose: Compare service reliability, facility detention and revenue efficiency across routes and facilities.

The page contains three principal visuals:

- On-Time Delivery by Route.
- Average Detention by Facility.
- Revenue per Mile by Route.

The route visuals highlight differences in delivery reliability and revenue generated relative to distance travelled. The facility visual identifies locations with comparatively high or low average detention. The principal insight is that delivery reliability is weak across the network, while route and facility differences provide opportunities for more targeted investigation. The results do not establish that detention alone explains late deliveries. This page supports follow-up analysis of route scheduling, facility processes and commercial performance.

Page 3 - Fleet & Driver Performance:

Purpose: Examine fuel expenditure, maintenance costs and recorded fleet utilisation.

The page contains three principal visuals:

- Monthly Fuel Expenditure.
- Top 10 Trucks by Maintenance Cost.
- Top 10 Trucks by Average Utilisation Rate.

The monthly expenditure trend shows how recorded fuel spending changes over time. The maintenance visual identifies trucks with high total maintenance costs, while the utilisation visual highlights trucks with the highest recorded average utilisation rates. Displaying the two truck rankings side by side makes it easier to identify vehicles that may warrant further investigation across both measures.

The principal insight is that fuel expenditure declined substantially between 2022 and 2024, while recorded fleet MPG improved only marginally. The maintenance and utilisation visuals provide a starting point for identifying vehicles requiring closer review, but further analysis is needed to establish the relationship between usage, cost and reliability. The page should be interpreted as a fleet investigation tool rather than a definitive measure of truck profitability or operational efficiency.

RECOMMENDATIONS

The findings support the following priorities.

Improve delivery reliability

Investigate the operational causes of the 55.39% recorded late delivery rate. Review planned and actual arrival times, route characteristics, scheduling practices, detention and facility processes. Establish a consistent on-time delivery baseline and monitor performance after interventions.

Investigate detention at higher-performing and lower-performing facilities

Begin with facilities reporting the highest average detention, including FAC00004, Chicago Distribution Center. Compare their operating processes, appointment practices and service characteristics with lower-detention facilities. Confirm that differences in facility function and delivery volume are accounted for before adopting changes.

Monitor revenue efficiency alongside costs

Continue monitoring revenue per mile as an indicator of revenue generation relative to distance travelled. Where the necessary data is available, combine revenue with fuel, maintenance, driver, insurance, toll and other operating costs to develop a more complete measure of route and vehicle profitability.

Separate fuel expenditure from fuel efficiency

Monitor fuel spending, cost per gallon, gallons consumed and MPG as distinct measures. Investigate truck-level fuel efficiency and consider driver coaching, reduced idling and preventive maintenance where the data supports these interventions. Evaluate any improvements using consistent measurement periods and validated mileage and fuel-consumption records.

Prioritise maintenance investigations using multiple indicators

Investigate trucks with high maintenance expenditure, frequent maintenance events or unusually high average repair costs. Compare these measures with vehicle age, mileage, utilisation and downtime. Avoid making replacement decisions based on maintenance expenditure alone.

Validate utilisation metrics

Confirm the calculation and expected range of the supplied utilisation rate before using it as an operational target. Investigate records above 100% and establish whether they reflect the metric's design or data quality problems. Once validated, use utilisation alongside mileage, maintenance costs and service performance to support fleet allocation decisions.

Strengthen operational data quality

Improve the completeness of truck, driver and trailer identifiers in trip records. Investigate the repeated maximum detention value and ensure that reporting-period filters are applied consistently. These improvements would strengthen the reliability of future dashboards and enable more precise analysis of individual vehicles, routes, facilities and drivers.

LIMITATIONS

The findings must be interpreted in the context of the supplied data and the scope of the analysis.

Incomplete operating costs: The dataset does not include all costs required to calculate true profit. Revenue per mile is therefore a revenue efficiency measure rather than a profitability measure.

Missing identifiers: Some trip records lack truck, driver or trailer identifiers, limiting the completeness of vehicle- and driver-level analysis.

Utilisation metric uncertainty: Some truck utilisation records exceed 100%. The meaning and validity of these values require confirmation before the measure is used for operational targets.

Detention data characteristics: All facilities report a maximum detention of 239 minutes. The reason for this uniform maximum is not confirmed.

Fuel efficiency attribution: Changes in fuel expenditure cannot be attributed solely to operational efficiency because recorded fuel prices also changed. The annual MPG figures show only a marginal improvement.

Descriptive analysis: The project identifies patterns and potential relationships but does not establish causation. Additional analysis is needed before concluding that a particular route, facility, utilisation level or maintenance pattern causes an operational outcome.

Reporting period: The principal reporting period is January 2022 to December 2024. Some records extend into January 2025 and must be handled consistently.

Currency denomination: The dataset does not establish a confirmed currency denomination. Financial values are therefore presented in recorded currency units.

Dataset representativeness: The findings describe the supplied dataset and should not automatically be generalised to a real-world logistics organisation without additional context and validation.

CONCLUSION

This project demonstrates how relational data analysis and business intelligence visualisation can be applied to logistics and fleet performance questions.

The analysis identified a significant service reliability concern, with 44.61% of deliveries recorded as on time and average delivery detention of 106.60 minutes. Route and facility comparisons highlighted areas for further investigation, while revenue-per-mile analysis provided an initial view of commercial performance.

Fuel analysis showed that recorded expenditure declined by approximately 12.4% between 2022 and 2024, alongside a decline in recorded cost per gallon and broadly stable consumption. Fleet MPG improved by only approximately 0.14%, suggesting that lower recorded fuel prices were more significant than measured efficiency gains in explaining the reduction in fuel expenditure.

Maintenance and utilisation analysis highlighted trucks requiring further investigation and established a framework for comparing vehicle-related measures. However, additional mileage, downtime, vehicle age and validated utilisation data are needed to establish a stronger relationship between vehicle use, maintenance expenditure and reliability.

The project also identified data quality issues that affect operational interpretation, including missing vehicle and driver identifiers, utilisation records above 100% and a uniform maximum detention value across facilities.

Overall, the project demonstrates a structured analytical workflow from data validation and SQL investigation through to KPI development, dashboard design and evidence-based recommendations.

The resulting dashboard provides a foundation for operational decision support. With improved data quality and additional cost and vehicle-level information, the analysis could be extended to support more detailed fleet efficiency, route profitability and service improvement investigations.